Langchen Liu

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Education

Yale University — **Sep 2023 – *May 2027 (expected)***
Ph.D., Statistics & Data Science — *New Haven, CT*

University of Pennsylvania — **Aug 2022 – Jul 2023 (transferred)**
Ph.D. (first year), Applied Math & Computational Science; GPA: 4.0/4.0 — *Philadelphia, PA*

Duke University — **Aug 2018 – May 2022**
B.S., Applied Mathematics; Major GPA: 3.98/4.0 — *Durham, NC*

Work Experience

Schonfeld Strategic Advisors — **Jun 2026 – Aug 2026**
Quantitative Research Intern, Neutrality Team — *New York, NY*

- **Ownership:** took over a portfolio optimization project recommended for production and originally scoped beyond the internship; ramped into the production optimizer/kdb+/MOSEK stack in week **1**, delivered core results within **4** of **10** weeks, and presented results **weekly** to the **16**-person research team.
- **New method:** designed a novel two-step portfolio construction from risk-space geometry: proved the high-dimensional risk ellipsoid is extremely flat, then used its eigenvalues to trim the ill-conditioned alpha vector into a well-conditioned one.
- **Impact:** at the same risk budget and GMV, raised 2023–2025 backtested net PnL by **3%** while shifting PnL from factor to idiosyncratic alpha and cutting out-of-budget factor loadings by ∼**85%**: more alpha with less factor risk.
- **Market-neutral projection:** designed a projection method on the Barra factor model that finds the portfolio orthogonal to the market factor, reducing total portfolio risk by **60%**.
- **AI tooling:** built Claude Code/Codex agent workflows for QR data exploration, code review, and parallel experiment sweeps; presented to the same team and cut data-analysis time by ∼**90%** with reusable workflow templates.

Selected Research

FoundPDE: Generative Foundation Model for Symbolic PDE Discovery — **Jul 2024 – Present**
Co-first Author (with Junlin Luo); Advisor: Lu Lu — *Yale University*

- **One model, many equations:** built a Transformer that reads spatiotemporal fields and writes the governing PDE as symbols (Fourier/derivative features, BFGS coefficient refinement), replacing per-dataset symbolic search with one pretrained model.
- **Accuracy and generalization: 93.4%** type recognition across **20** PDE families (95% CI [**91.8%, 94.8%**]) and **89.4%** under out-of-range coefficient shift; few-shot fine-tuning exceeded **90%** exact recovery on **4** unseen composite families within **20** epochs.
- **Evaluation rigor:** audited the pipeline end to end, found and removed a hidden evaluation-time noise default that contaminated "clean" results, separated coefficient-shift from noise effects, and added clustered bootstrap confidence intervals.

PDEDiscoveryBench: Evaluating Differential Equation Discovery from Spatiotemporal Data — **2026 – Present**
First Author; under review at ICLR 2027 — *Yale University*

- **Full-workflow evaluation:** built **337** ODE/PDE systems (**332** with reference equations, **5** unlabeled CFD blood-flow fields) at **3** prior levels (generic to equation-informed) and **4** noise tiers, scoring accuracy, noise robustness, compute cost, and pipeline failures for **7** methods: SINDy, WSINDy, SGA-PDE, DISCOVER, LLM-PDESR, and GPT-5.6-sol and Claude Opus 5 agents.
- **Findings:** equation-informed priors beat generic ones in exact recovery at every noise tier for all **6** noise-tested methods; with physics hidden, Claude agents proposed Stokes balance on all **5** flow fields and Navier-Stokes momentum on the **2** highest-Re ones.

PDEScientist: An AI Researcher for Equation Discovery — **2026 – Present**
Anthropic AI for Science Program award (Oct 2026); with Lu Lu (PI), Yinling Zhang — *Yale University*

SciExam for ENSO: Can AI Agents Build Climate Models? — **2026**
Second Author — *Yale University*

- **Depth over a single metric:** **12** agent systems (Claude Code, Codex, Antigravity) build stochastic ENSO models from real data in **6** hours; hidden graders score statistics, reconstruction, and forecasts. **6/12** beat a published reference model (best **+36%**); their simplified forms fall into two families matching rival expert explanations of ENSO asymmetry, a debate the task never mentions.

Feasibility of Federated Learning in Scientific Machine Learning (FedSciML) — **Sep 2022 – May 2024**
Coordinator: Handi Zhang, Lu Lu — *Yale University*

- **Theory and experiments:** proved an **O(local epochs)** bound on federated-vs-global weight divergence under heterogeneous data; ran **500+** experiments on **10+** functions/PDEs incl. Navier-Stokes (avg. **L2 error within 5%**).

Multicultural Emotion in Multilingual Language Models — **Dec 2022 – Apr 2023**
Coordinator: Shreya Havaldar, Sunny Rai, Bhumika Singhal, Lyle Ungar — *University of Pennsylvania*

- **Open-source pipeline:** built the embedding analysis pipeline for **9,000+** emotion-language-model projections across **15+** languages and **30+** models; co-authored ACL WASSA 2023 paper with **123** citations.

Technical Skills

- **Programming:** Python (NumPy, pandas, PyTorch, scikit-learn), C++, q/kdb+, Julia, R, SQL, Java
- **ML/Stats:** Statistical learning, Bayesian inference, time series, cross-validation, regularization, ablation design
- **AI/Agents:** Claude Code, Codex, multi-agent coding workflows, agent skill design, tool/API orchestration
- **Systems/Tools:** Git, Docker, Slurm/HPC, experiment tracking, pytest, HDF5/Parquet
- **Math:** Probability, stochastic processes, measure theory, functional analysis, convex optimization, numerical linear algebra